Supplementary Table 1 Supplementary data

Author, [ref.] year	Study cohorts	(N) per cohort	Underlying disease							Location of Impaction		
			Peptic stricture	Eosinophilic esophagitis	Schatzki ring	Anastomotic stricture	Tumor	No pathology (normal esophagus)	Other	Distal	Middle	Proximal
Ooi 2021 [6]	Cap-assisted	171	69	48	8	6	0	40	NR	130	35	6
	Conventional	171	63	51	9	4	0	44	NR	132	37	2
Ooi 2018 [7]	Cap-assisted	93	26	18	5	4	1	39	NR	61	27	5
	Conventional	106	38	24	5	3	3	33	NR	71	31	4
Wahba 2019 [8]	Cap-assisted	106	21	4	5	1	4	37	post corrosive esophagitis-32; esophagitis-2	23	37	46
	Conventional	110	24	6	5	4	6	34	post corrosive esophagitis-30; esophagitis-1	27	34	49
Zhang 2013 [9]	Cap-assisted	35	0	0	0	0	0	0	1 (hiatal hernia), 2 (peptic ulcer); esophagitis-3	0	0	35
	Conventional	35	0	0	0	0	0	0	2 (hiatal hernia), 1 (peptic ulcer), 1 (achalasia); esophagitis-2	0	0	35
Fang 2020 [5]	Cap-assisted	224	0	0	0	0	2	209	4; benign stricture-9	24	67	133
	Conventional	224	0	0	0	0	13	164	11; benign stricture-36	26	69	129
Zhao 2017 [10]	Cap-assisted	48	NR	NR	NR	NR	NR	NR	NR	NR	NR	NR
	Conventional	48	NR	NR	NR	NR	NR	NR	NR	NR	NR	NR

NR, not reported

Supplementary Table 2A Study quality assessment – Jadad scoring system

Author, [ref.] year	Randomization			Blinding			Account of all patients	Score (total=5)	Risk of bias
	Randomization mentioned: +1	Randomization appropriate: +1	Inappropriate method of randomization: -1	Blinding mentioned: +1	Method appropriate: +1	Method inappropriate: -1	All pts accounted for: +1		
Ooi 2021 [6]	1	1	0	0	0	-1	1	2	Moderate
Zhang 2013 [9]	1	1	0	0	0	-1	1	2	Moderate
Zhao 2017 [10]	1	1	0	0	0	-1	1	2	Moderate

Supplementary Table 2B Study quality assessment – Newcastle-Ottawa scoring system

Author, [ref.] year	Selection				Comparability	Outcome			Score	Quality
	Representativeness of the average adult in community	Cohort size	Information on outcomes	Outcome not present at start	Comparison group adequately controlled for	Adequate clinical assessment	Follow-up time	Adequacy of follow up	MAX=8, HIGH>6, MEDIUM 4-6, LOW<4	
	Population based: 1; Multi-center: 0.5; Single-center: 0	>40 patients: 1; 39-20: 0.5; <20: 0	Information with clarity: 1; information derived: 0.5	Not present: 1; present: 0	Yes: 1; no: 0	Yes: 1; no: 0	Yes: 1; not mentioned: 0	All patients followed up: 1; >50% followed up: 0.5; <50% followed up OR not mentioned: 0		
Ooi 2018 [7]	0	1	1	1	1	1	1	1	7	High
Wahba 2019 [8]	0.5	1	1	1	1	1	1	1	7.5	High
Fang 2020 [5]	0	1	1	1	1	1	1	1	7	High

Appendix

Searches ran on 02/02/2022

OVID

Database(s): Ovid MEDLINE® ALL (1946 to February 02, 2022), EBM Revie–s - Cochrane Central Register of Controlled Trials February 2022, EBM Revie–s - Cochrane Database of Systematic Reviews

Appendix 1 Literature search strategy

#	Searches	Results
1	*upper endoscopy/or *EGD/or esophagoscopy or *esophagogastroduodenoscopy	204
2	*food bolus/or *food impaction/or *foreign body/or food bolus.mp. or foreign body.mp. or food impaction.mp.	371
3	*clear cap/or *cap assisted endoscopy/or cap.mp. or clear cap.mp.	475
4	1 and 2 and 3	1050
5	remove duplicates from 4	704

PubMed, 1816 results (English only)

((endoscopy [majr] AND cap-assisted [majr]) “R “”GD”[majr] “R “clear ”ap”[majr] OR esophagogastroduodenoscopy [tiab]) AND“
(“food bo”us”[majr] “R “foreign b”dy”[majr] OR food impaction [tiab] OR foreign body impaction [tiab]) AND“
(“upper endosc”py”[majr] OR esophagogastroduodenoscopy [tiab])

Scopus

1	TITLE-ABS-KEY (“("endosc”py" O“ ""GD" OR esophagogastroduodenoscopy) AND “("Food bo”us" O“ "foreign b”dy" O“ "food impact”on" O“ "foreign body impact”on") AND “("clear ”ap" O“ "cap assis”ed" OR cap) AND (endoscopy))	105

Web of Science

“	("endosc”py" O“ ""GD" OR esophagogastroduodenoscopy) AND “("Food bo”us" O“ "foreign b”dy" O“ "food impact”on" O“ "foreign body impact”on") AND “("clear ”ap" O“ "cap assis”ed" OR cap) AND (endoscopy)	164

973 total article references
258 duplicates found in EndNote
715 total references in EndNote

Appendix 2 MOOSE checklist [11]

Item No	Recommendation	Reported on Page No
	Reporting of background should include	
1	Problem definition	5
2	Hypothesis statement	-na-
3	Description of study outcome (s)	5-6
4	Type of exposure or intervention used	5-6
5	Type of study designs used	-na-
6	Study population	5-6
	Reporting of search strategy should include	
7	Qualifications of searchers (e.g., librarians and investigators)	7, Appendix 1
8	Search strategy, including time period included in the synthesis and key words	7, Appendix 1
9	Effort to include all available studies, including contact with authors	8
10	Databases and registries searched	Appendix 1
11	Search software used, name and version, including special features used (e.g., explosion)	-na-
12	Use of hand searching (e.g., reference lists of obtained articles)	8
13	List of citations located and those excluded, including justification	Figure 1
14	Method of addressing articles published in languages other than English	8
15	Method of handling abstracts and unpublished studies	7, 8
16	Description of any contact with authors	8
	Reporting of methods should include	
17	Description of relevance or appropriateness of studies assembled for assessing the hypothesis to be tested	8
18	Rationale for the selection and coding of data (e.g., sound clinical principles or convenience)	8
19	Documentation of how data were classified and coded (e.g., multiple raters, blinding and interrater reliability)	-na-
20	Assessment of confounding (e.g., comparability of cases and controls in studies where appropriate)	-na-
21	Assessment of study quality, including blinding of quality assessors, stratification or regression on possible predictors of study results	8, Supplementary Table 2A, B
22	Assessment of heterogeneity	9
23	Description of statistical methods (e.g., complete description of fixed or random effects models, justification of whether the chosen models account for predictors of study results, dose-response models, or cumulative meta-analysis) in sufficient detail to be replicated	9
24	Provision of appropriate tables and graphics	provided
	Reporting of results should include	
25	Graphic summarizing individual study estimates and overall estimate	Forest plots, Table 2
26	Table giving descriptive information for each study included	Table 1
27	Results of sensitivity testing (e.g., subgroup analysis)	Table 2
28	Indication of statistical uncertainty of findings	13

Item No	Recommendation	Reported on page No
	Reporting of discussion should include	
29	Quantitative assessment of bias (e.g., publication bias)	13
30	Justification for exclusion (e.g., exclusion of non-English language citations)	-na-
31	Assessment of quality of included studies	Supplementary Table 2
	Reporting of conclusions should include	
32	Consideration of alternative explanations for observed results	14-16
33	Generalization of the conclusions (i.e., appropriate for the data presented and within the domain of the literature review)	14
34	Guidelines for future research	16
35	Disclosure of funding source	17

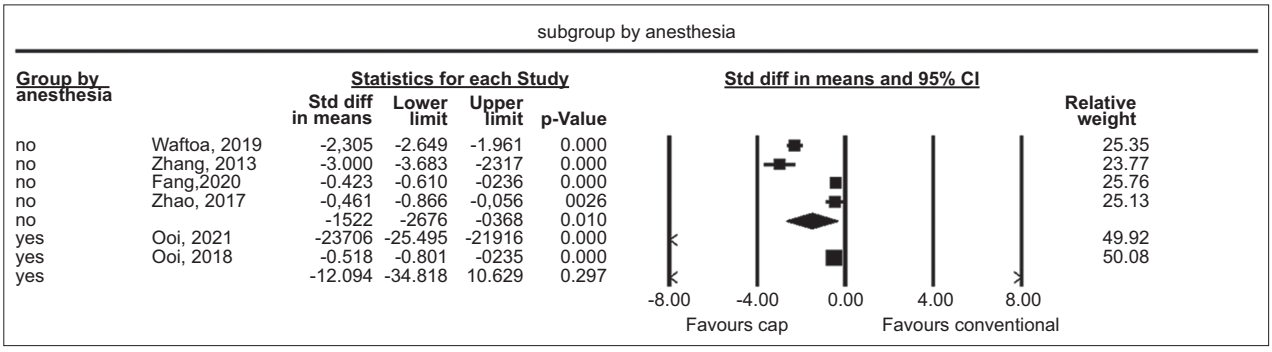
Appendix 3 PRISMA checklist

Section/topic	#	Checklist item	Reported on page #
		Title	
Title	1	Identify the report as a systematic review, meta-analysis, or both	1
		Abstract	
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number	4
		Introduction	
Rationale	3	Describe the rationale for the review in the context of what is already known	5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS)	6
		Methods	
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number	-na-
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale	7
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched	Appendix-1
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated	8, appendix-1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis)	8
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators	8
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made	8
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis	9
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means)	9
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis	9
Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies)	11
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified	12
		Results	
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram	10, 11, 12
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations	10, Table 1
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12)	11, Supplementary Table 2
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot	11, 12

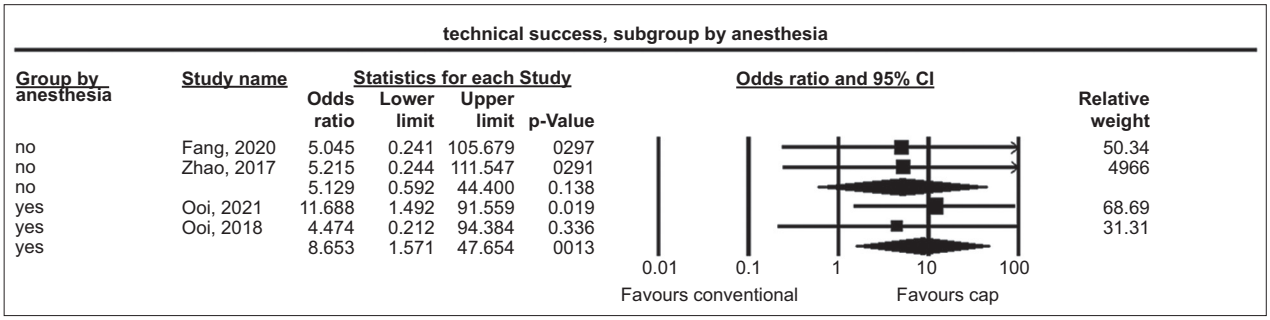
(Contd...)

Appendix 3 *(Continued)*

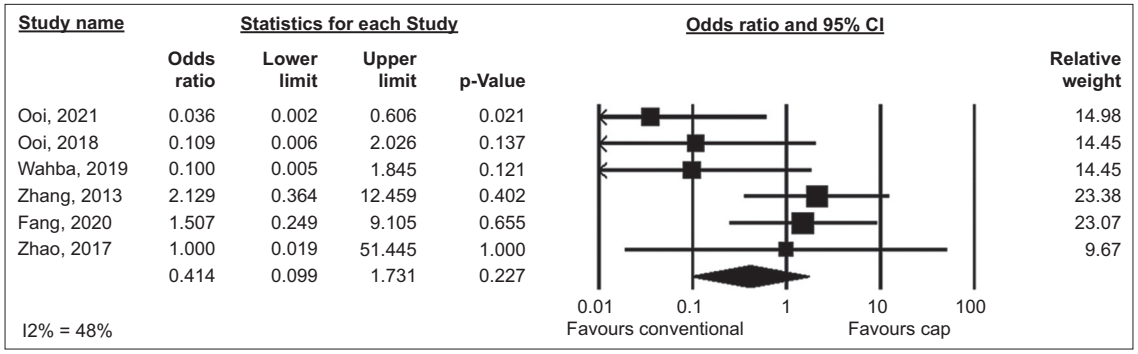
Section/topic	#	Checklist item	Reported on page #
Synthesis of results	21	Present the main results of the review. If meta-analyses are done, include for each, confidence intervals and measures of consistency	11, 12
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15)	Supplementary Table 2
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16])	13, Table 2
		Discussion	
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers)	14-16
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias)	16
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research	16
		Funding	
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review	17



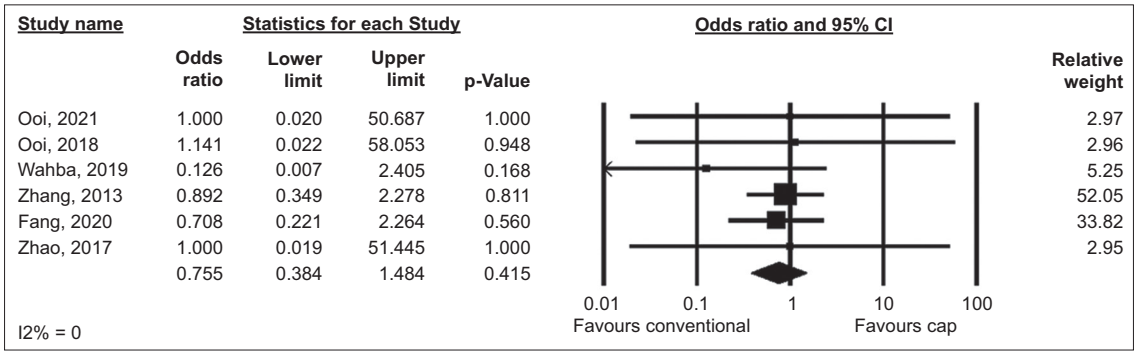
Supplementary Figure 1 Forest plot, subgroup analysis of technical success by anesthesia
CI, confidence interval



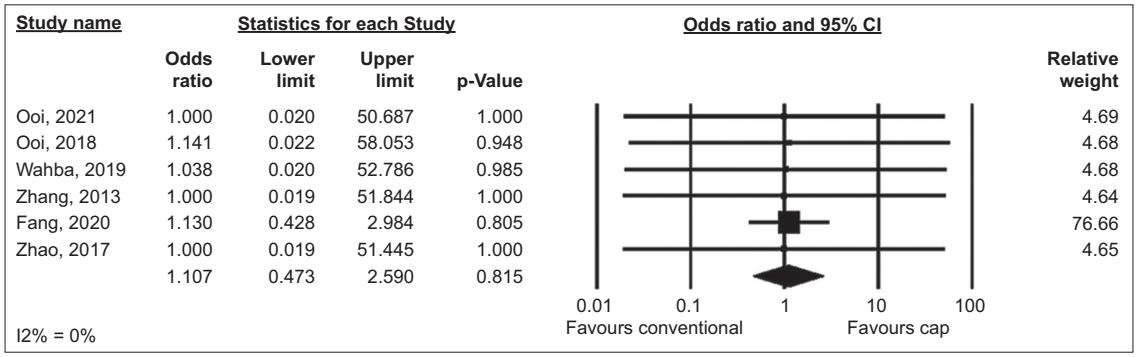
Supplementary Figure 2 Forest plot, subgroup analysis of procedure time by anesthesia
CI, confidence interval



Supplementary Figure 3 Forest plot, mucosal tear
CI, confidence interval



Supplementary Figure 4 Forest plot, locoregional bleeding
CI, confidence interval



Supplementary Figure 5 Forest plot, perforation
CI, confidence interval